

# Supplementary Information for “Validation-Aligned Coreset Selection for Budgeted Few-Shot Classification”

## **Supplementary Note 1: claim boundary**

The main article reports a bounded claim. VACS-R is the strongest evaluated training-only deployable protocol in the five-dataset low-budget 3-nearest-neighbour benchmark, but the experiments do not establish broad state-of-the-art performance. The supplementary tables below document the paired fast-protocol audit, ablations, non-deployable hindsight-static reference, validation-size diagnostics, selection costs, and frozen embedding boundary checks.

## **Supplementary Table 1: paired low-budget fast-protocol stability**

This table reports VACS-F minus herding differences over matched low-budget units.

| Slice         | $n$ | $\Delta$ Acc. | 95% CI        | W/T/L    | Sign $p$ |
|---------------|-----|---------------|---------------|----------|----------|
| Pooled        | 80  | +0.02         | [-1.38,+1.20] | 28/41/11 | 0.009    |
| 20NG          | 16  | +4.54         | [+2.43,+7.07] | 16/0/0   | < .001   |
| Breast cancer | 16  | +0.84         | [+0.37,+1.32] | 7/9/0    | 0.016    |
| Digits        | 16  | +0.19         | [-0.12,+0.53] | 3/12/1   | 0.625    |
| Iris          | 16  | -4.31         | [-9.86,-0.14] | 2/9/5    | 0.453    |
| Wine          | 16  | -1.16         | [-2.20,-0.35] | 0/11/5   | 0.062    |

## Supplementary Table 2: ablation diagnostics

The ablation table separates the validation wrapper from individual selector candidates and from the final rebuild convention.

| Variant              | Acc. | $\Delta$ | Macro-F1 | Selected |
|----------------------|------|----------|----------|----------|
| Full VACS            | 70.6 | +0.00    | 70.2     | 11.4     |
| No final rebuild     | 72.0 | +1.34    | 71.7     | 11.4     |
| No K-means candidate | 70.4 | -0.21    | 69.8     | 11.4     |
| No MARC candidate    | 70.7 | +0.08    | 70.3     | 11.4     |
| No random candidate  | 70.6 | -0.05    | 70.3     | 11.4     |

## Supplementary Table 3: non-deployable dataset-hindsight static audit

The per-dataset static reference is selected with hindsight from the low-budget result table, and is therefore diagnostic rather than deployable.

| Slice                   | Static ref. | VACS | Static | $\Delta$ [95% CI]   | W/T/L   |
|-------------------------|-------------|------|--------|---------------------|---------|
| Pooled per-dataset-best | Per-dataset | 70.6 | 72.3   | -1.64 [-2.98,-0.61] | 8/53/19 |
| 20NG                    | Random      | 9.7  | 12.1   | -2.49 [-5.33,-0.03] | 4/7/5   |
| Breast cancer           | K-means     | 93.4 | 93.5   | -0.15 [-0.37,+0.00] | 0/14/2  |
| Digits                  | K-means     | 82.3 | 82.4   | -0.12 [-0.50,+0.19] | 2/12/2  |
| iris                    | Herding     | 76.4 | 80.7   | -4.31 [-9.86,-0.14] | 2/9/5   |
| Wine                    | Herding     | 91.3 | 92.5   | -1.16 [-2.20,-0.35] | 0/11/5  |

## Supplementary Table 4: validation-size reliability

This diagnostic repeats the internal validation-ranking analysis at 10%, 20%, 25%, and 33% validation fractions.

| Val. frac. | Units | VACS | Gain  | Top-1 | $\rho$ | Regret | Severe |
|------------|-------|------|-------|-------|--------|--------|--------|
| 0.10       | 64    | 69.1 | +0.94 | 60.9  | 0.81   | 1.58   | 12.5   |
| 0.20       | 64    | 68.3 | +0.10 | 46.9  | 0.77   | 2.42   | 18.8   |
| 0.25       | 64    | 69.3 | +1.13 | 60.9  | 0.81   | 1.38   | 9.4    |
| 0.33       | 64    | 69.0 | +0.86 | 59.4  | 0.80   | 1.66   | 14.1   |

## Supplementary Table 5: selection cost

The cost diagnostic reports selection-only time for herding and validation-plus-rebuild time for VACS-F over the matched low-budget decisions.

| Slice         | $n$ | Herding | VACS val. | Rebuild | Total  |
|---------------|-----|---------|-----------|---------|--------|
| Pooled        | 80  | 10.475  | 79.820    | 7.390   | 87.210 |
| 20NG          | 16  | 10.250  | 75.500    | 6.859   | 82.359 |
| Breast cancer | 16  | 0.038   | 0.969     | 0.107   | 1.076  |
| Digits        | 16  | 0.137   | 2.011     | 0.336   | 2.347  |
| Iris          | 16  | 0.018   | 0.667     | 0.038   | 0.705  |
| Wine          | 16  | 0.032   | 0.673     | 0.050   | 0.723  |

## Supplementary Table 6: frozen transformer probes

The frozen transformer probes are boundary evidence. They do not support a broad modern text-embedding superiority claim.

| Probe                    | Dim. | VACS | Best static  | Gain  |
|--------------------------|------|------|--------------|-------|
| DeBERTa-v3-small         | 768  | 9.8  | Herding 9.9  | -0.17 |
| SmolLM2-360M, maxlen 64  | 960  | 31.5 | MARC 31.4    | +0.03 |
| SmolLM2-360M, maxlen 128 | 960  | 37.4 | Herding 37.5 | -0.12 |
| Qwen2.5-0.5B             | 896  | 40.1 | Herding 40.2 | -0.11 |

## Supplementary Tables 7 and 8: frozen image-embedding probes

The image probes widen benchmark coverage but tie the strongest static selector in the tested one-seed, one-example-per-class settings.

| Dataset       | VACS | Best static  | Gain  | Backbone      |
|---------------|------|--------------|-------|---------------|
| Fashion-MNIST | 75.6 | Herding 75.6 | +0.00 | convnext_tiny |
| Pooled        | 75.6 | Herding 75.6 | +0.00 | convnext_tiny |

| Dataset  | VACS | Best static  | Gain  | Backbone |
|----------|------|--------------|-------|----------|
| CIFAR-10 | 64.7 | Herding 64.7 | +0.00 | resnet18 |
| Pooled   | 64.7 | Herding 64.7 | +0.00 | resnet18 |

## Supplementary Table 9: full 20-class sparse-text stress boundary

The compact full-20-class TF-IDF stress check is negative for VACS relative to static random and is kept as a claim boundary.

| Task                     | Classes | Units | VACS | Best static | Gain  | Top VACS choice |
|--------------------------|---------|-------|------|-------------|-------|-----------------|
| 20 Newsgroups (20-class) | 20      | 16    | 9.7  | Random 12.1 | -2.49 | Boundary 9/16   |

## Supplementary Note 2: reproducibility files

The project package includes the Python scripts used to regenerate the primary result manifest, all tables included in the manuscript and supplementary information, the workflow and diagnostic

1 figures, and the consistency checks that compare manuscript claims with generated result files. The  
2 main experiment driver is `experiments/run_vacs_experiments.py`. Covertypes audits,  
3 repeated-validation audits, validation-size diagnostics, and figure generation are implemented as  
4 separate companion scripts for auditability.